Simulating Urban Evacuation Traffic Under Authoritative Hurricane Forecasting with Agent-Based Modeling

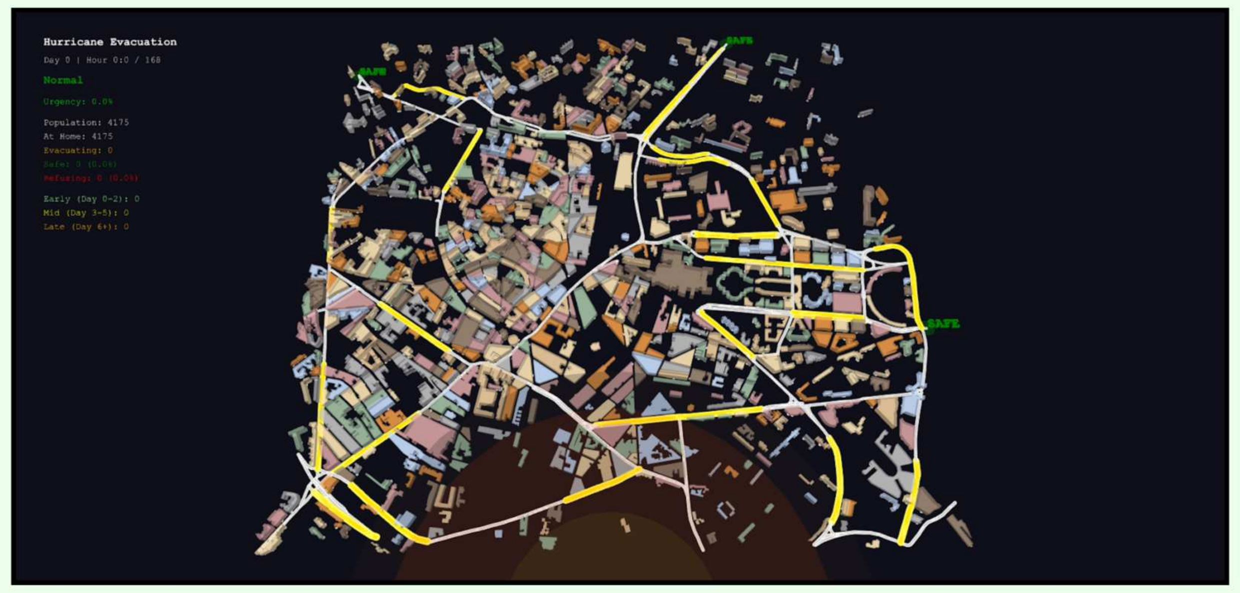
Introduction

Problem:
During emergencies, namely hurricanes, urban populations exhibit unpredictable behavior in traffic as they are actively evacuating in response to evolving forecasts. Being geographically anchored, differing risk tolerance, social conformity, and stubbornness levels will lead to the convergence or divergence within group behaviors.

Significance:
Insights from this project are transferrable to other/future disasters, including wildfires, tornadoes, and floods. Beyond human behavior, it also explores how misinformation and social influence can either mitigate or exacerbate traffic in such crises. Urban systems can be improved upon through better emergency planning, public policy design, and government communication. That way, immediate cooperative action and trust can overcome these unfortunate disasters.

Background

An agent-based model (ABM) was used to simulate an urban population responding to an approaching hurricane. Widespread growth of ABMs have been seen due to the COVID-19 pandemic. By applying relatively simple properties, these models can display large-scale emergent behavior of individuals' decisions. For example, Helbing et al. (2000) illustrated how deep-instilled panic and herding behavior leads to deaths from being trampled and crushed. Following this underlying cause-and-effect relationship, Geographic Information Systems (GIS) were added to better simulate realistic traffic trajectories within urban cities. This combination is starting to become a necessity, as a study by Senanayake et al. (2024) emphasizes how even just pedestrian evacuations simulated by ABMs require more realism to actually have a meaningful impact on city infrastructure improvement. Furthermore, one of the main premises of this study is social conformity under uncertainty. Classic, notable studies are Solomon Asch's conformity experiments—individuals conform to group opinions even though they are very clearly incorrect (Asch, S. E., 1956)—and Stanley Milgram's obedience experiments—individuals will readily obey authoritative figures, even if it meant harming others (Milgram, S., 1963). These experiments lay the core groundwork for the psychological component, along with the traffic/evacuation logistics that will be explored simultaneously.



References

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Methods

SPATIAL DATA + CONFIGURATION

Inputs:
- Buildings (shapefile geometry)
- Road network (graph structure)
- Time parameters, population config
- Agent trait distributions
- Speed + evacuation settings

GLOBAL STATE

Hurricane Forecast
- Position, uncertainty cone
- Immediate/warning/watch zones
- Updates every 4 steps (7-day track)
- Triggers govt announcements

Government System
- Announcement urgency level
- Triggers: watch/warning/mandatory
- Decay rate: 0.95/step
- Announcements: Days 0,2,3,4,5,6

Road Network + Traffic Field
- Road graph, base speeds per road type
- Speed coefficients (congestion-dependent)
- Traffic density field (150×150 grid)
- Updates every 2 steps; decay 0.7/step

PEOPLE/VEHICLE AGENTS

Static Traits:
risk_tolerance | social_conformity | preparedness
stubbornness | info_access

Dynamic State:
evacuation_status | stress_level | fatigue
perceived_risk (cached) | location, target

Decision Process (Every Step):

Risk Assessment:
Combine forecast position, zones, info_access -> perceived_risk (cached every 4 steps)

Social Pressure:
Count neighbors evacuating within 150m -> social_pressure (checked every 4 steps)

Evacuation Decision:
If (risk + social_pressure + urgency_effect + stress_effect) > personal_threshold -> evacuate

INFRASTRUCTURE AGENTS

Building Agents:
- Static geometry + visualization
- Track contained population

Road Agents:
- road_type (highway/local)
- base_speed_kmh, current_nb_people
- Dynamic speed_coeff (updated every 2 steps)

Methods (Cont.)

CONNECTED PROCESSES

Congestion Feedback Loop:
Road nb_people -> speed_coeff
-> pathfinding weights -> slower evacuation

Traffic Field Dynamics:
Moving agents add density to grid
Decay (0.7) + diffusion (0.8) per step

STRESS + URGENCY EFFECTS

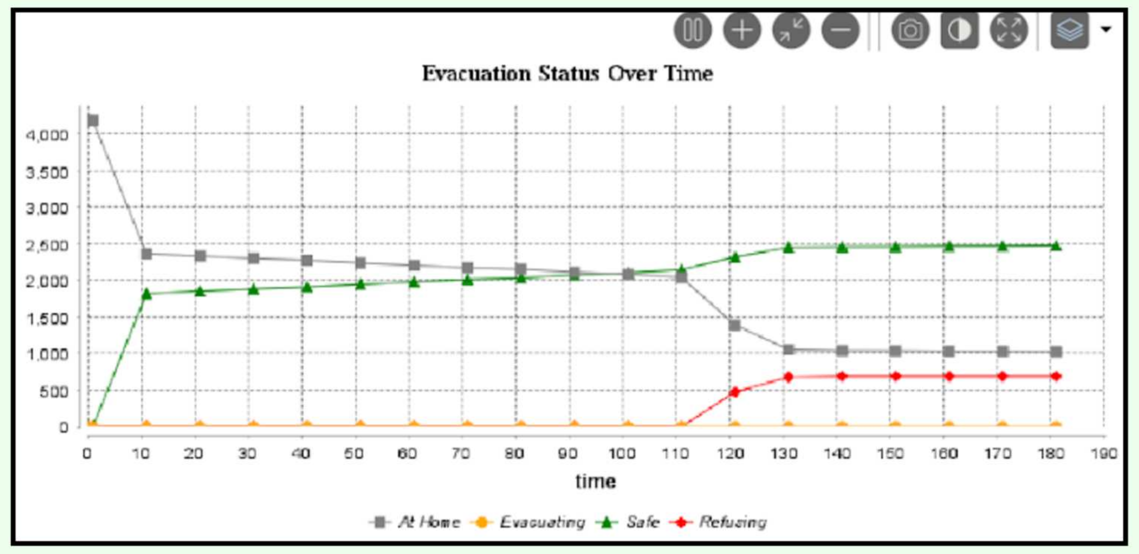
Stress Accumulation:
- Increases with: perceived_risk, social_pressure, urgency_level
- Fatigue reduces movement speed

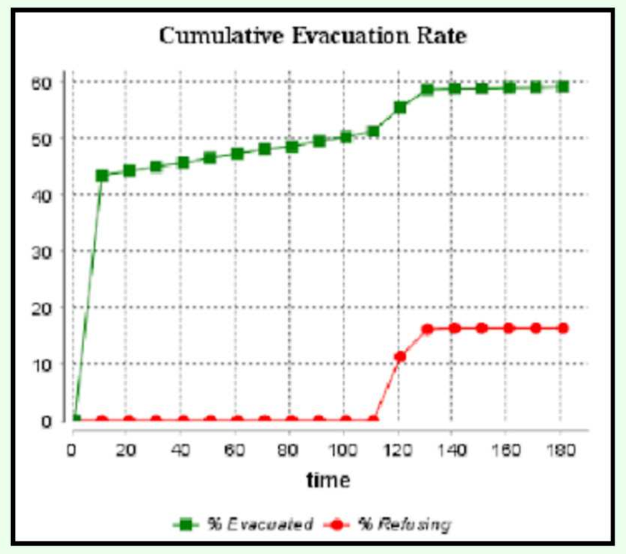
Urgency Effect (per announcement):
- Government announces trigger level
- Multiplies decision threshold temporarily
- Decays gradually (0.95/step)

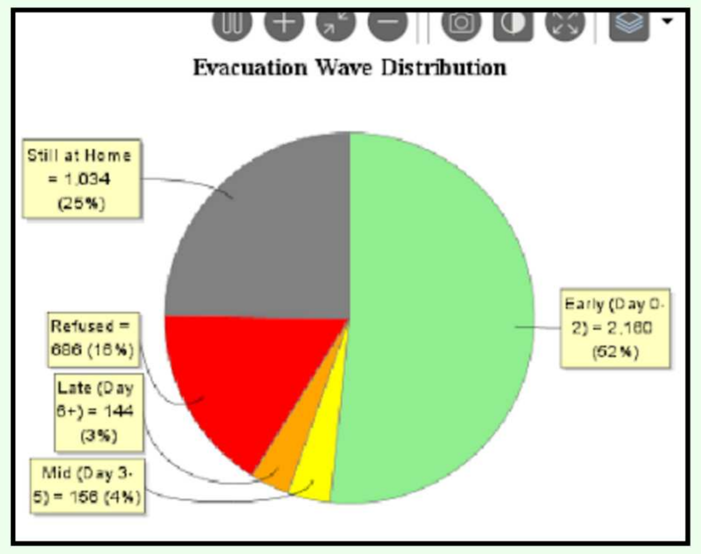
Emergent Behaviors:
- **Evacuation waves:** early, mid, late evacuators based on thresholds
- **Social contagion:** neighbors' decisions influence local clustering
- **Traffic jams:** congestion creates feedback that delays late evacuators

Results

The final ABM architecture allows for week-long scenarios to be simulated. It consists of seven major subsystems: (1) GIS data for building and road networks, (2) forecast tracking with time-related uncertainties, (3) heterogeneous decision making with six agent personality traits, (4) government announcements with three levels, (5) traffic congestion, (6) pathfinding for evacuation routing, and (7) data collection for various statistics.

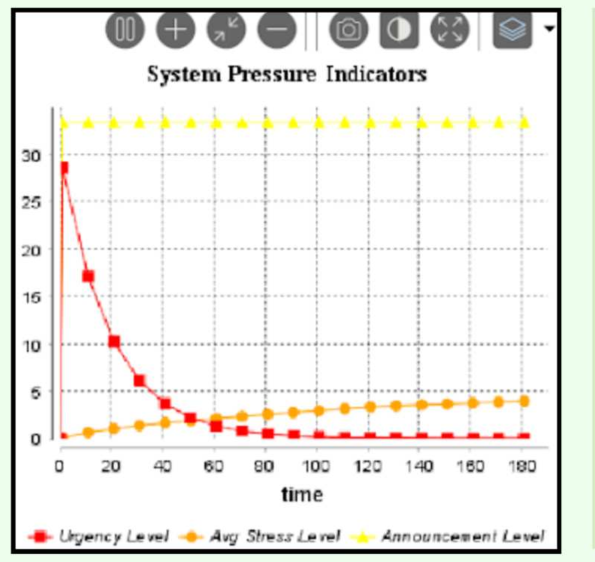






The simulation displays a clear evacuation pattern, as the population transitions between "At Home" and "Safe" from Hours 0-110 in the *Evacuation Status Over Time* graph. Afterwards, evacuation begins plateauing with 20% of agents refusing to leave or realizing it's too late, illustrating an overall successful mass evacuation, as seen in the *Cumulative Evacuation Rate* graph. Moreover, the pie chart *Evacuation Wave Distribution* portrays how approximately 50% of agents evacuated between Day 0-2 and less than 10% of agents evacuated between Day 3-6+, with the remaining either refusing to evacuate or still at home. This highlights the extent to which waves of evacuation are present among disaster scenarios.

Results (Cont.)



A major urgency drop from Hours 0-60 is visible, matching the decrease of initial panic response due to government announcements. The average stress level within agents gradually increases, illustrating the population's heterogeneity (some agents' traits make them prone to immediately evacuating without stress).

But, there exist agents who are not only geographically further from the hurricane's predicted forecast, but also have characteristics that set them on the fence.

Conclusions

The agent-based simulation produced valuable evacuation patterns. In the context of a week-long hurricane simulation, it highlights the importance of forecast uncertainty and social networks, which may be a meaningful insight for emergency planners. However, the current state of the project has significant room for advancements. Namely, due to computational limitations, synthetic geographic data was used to create *City A* where majority of testing was conducted. Similarly, sizing of real-world GIS data when extracting from BBBike for *City B* and *City C* had to be minimized. This prevented large-scale traffic dynamics from emerging, as vehicle speed had to be restricted. Therefore, future improvements include implementing large-scale/real-world cities, then comparing the model's results to historical data. On top of that, continuous verification and validation will reinforce confidence in the model's accuracy and credibility for real-world applications.

There would also be value in exploring other platforms to do this project on, such as Mesa or NetLogo. GAMA's built-in geographic data integration support expedites the process of designing and testing various shapefiles, but having a more abstract representation could allow analysis of specific components of an evacuation. For example, modeling a specific portion of a metro station, business district, or tourist landmark to see the inflow and outflow of people—not vehicles—as represented by artificial life: boids (Boids, n.d.).